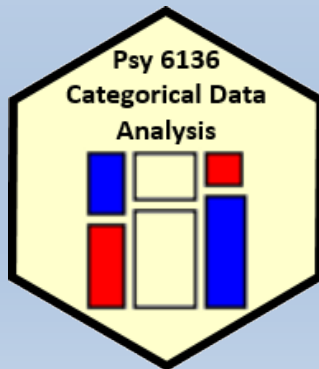




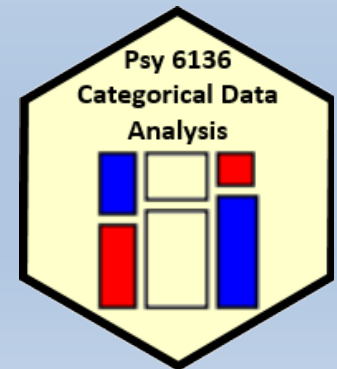
The Last Waltz

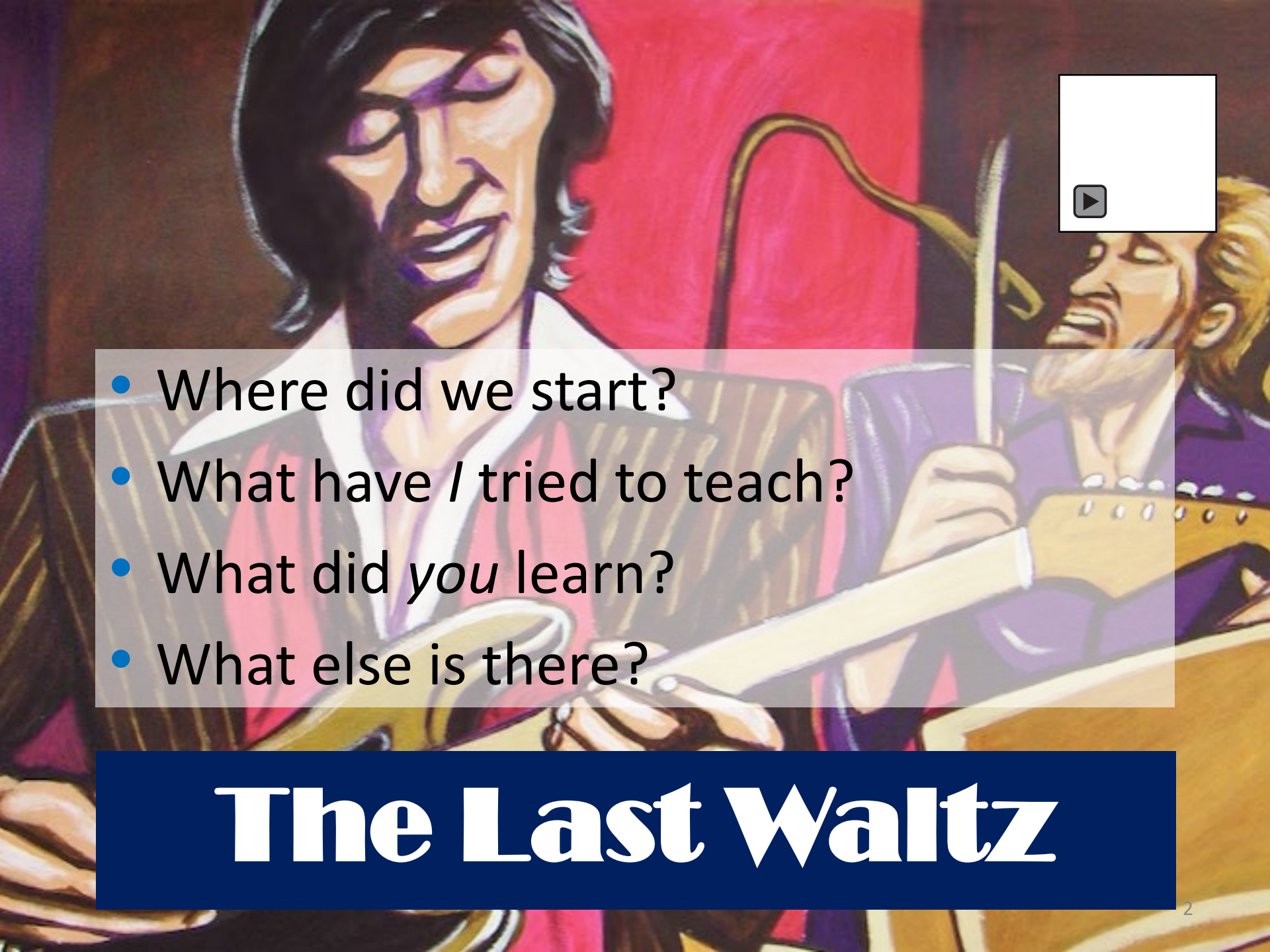
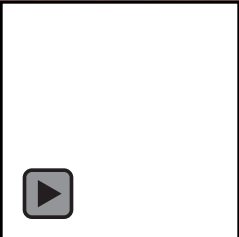


Michael Friendly

Psych 6136

<https://friendly.github.io/6136>



- 
- 
- Where did we start?
 - What have I tried to teach?
 - What did *you* learn?
 - What else is there?

The Last Waltz

My goals



Start with descriptive, hypothesis testing methods, then progress to model-based methods



Visual tools for thinking & understanding

Sieve plots, mosaic plots, spineplots,
...

Correspondence analysis: best 2D
summary

Effect plots, Data + Model plots



Build from simple, loglinear models to more complex ones

01: Overview

- Categorical data involves some new ideas
 - Discrete variables: `unordered` or `ordered`
 - Counts, frequencies as outcomes
- New / different data structures & functions
 - tables – 1-way, 2-way, 3-way, ... `table()`, `xtabs()`
 - similar in matrices or arrays `matrix()`, `array()`
 - datasets:
 - frequency form
 - case form
- Graphical methods: often use $\text{area} \sim \text{Freq}$
 - Consider: graphical comparisons, effect order
- Models: Most are $\cong$ natural extensions of `lm()`

Categorical data: Structures

Categorical (frequency) data appears in various forms

- **Tables:** often the result of `table()` or `xtabs()`

- 1-way
- 2-way – 2×2 , $r \times c$
- 3-way

Gender compared to handedness

	Handed		
	Left	Right	
Female	7	46	53
Male	5	63	68
	12	109	121

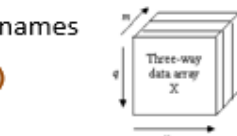
margins

- **Matrices:** `matrix()`, with row & col names

- **Arrays:** `array()`, with `dimnames()`

- **Data frames**

- Case form (individual observations)
- Frequency form



	Hair	Eye	Freq
1	Black	Brown	68
2	Brown	Brown	119
3	Red	Brown	26
4	Blond	Brown	7
5	Black	Blue	20
6	Brown	Blue	84
7	Red	Blue	17
8	Blond	Blue	94

Effect ordering: Frequency tables

- Effect ordering and high-lighting for tables

Table: Hair color - Eye color data: Effect ordered

Eye color	Hair color			
	Black	Brown	Red	Blond
Brown	68	119	26	7
Hazel	15	54	14	10
Green	5	29	14	16
Blue	20	84	17	94

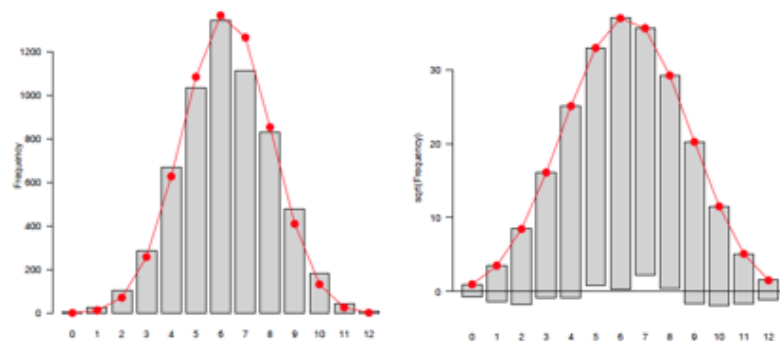
Model:	Independence: [Hair][Eye] $\chi^2(9) = 138.29$						
Color coding:	<-4	<-2	<-1	0	>1	>2	>4
n in each cell:	n < expected				n > expected		

The pattern is clearer when the eye colors are **permuted**: light hair goes with light eyes & vice-versa

1-way tables: graphs

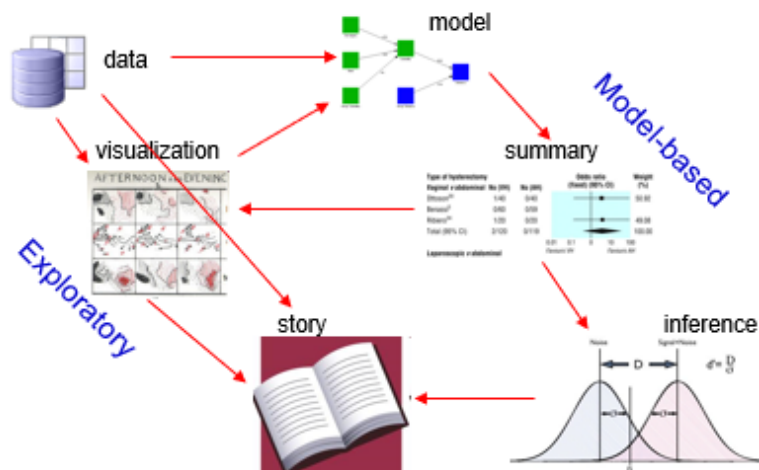
For a particular distribution in mind:

- Plot the data together with the fitted frequencies
- Better still: **hanging rootogram**: freq on sqrt scale; hang bars from fitted values



Data, pictures, models & stories

Now, tell the story!



02: Discrete distributions

- Discrete distributions are the building blocks for categorical data analysis
 - Typically consist of basic counts of occurrences, with varying frequencies
 - Most common: binomial, Poisson, negative binomial
 - Others: geometric, log-series
- Fit with `goodfit()`; plot with `rootogram()`
 - Diagnostic plots: `Ord_plot()`, `distplot()`
- Models with predictors
 - Binomial → logistic regression
 - Poisson → poisson regression; logliner models
 - These are special cases of **generalized** linear models

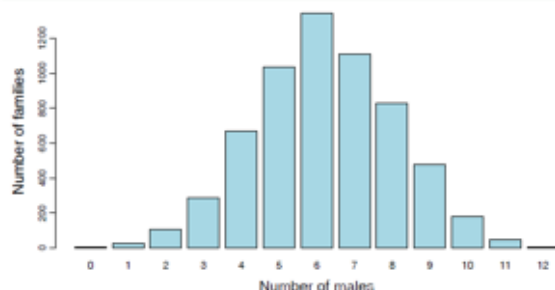
Examples: binomial

Human sex ratio (Geissler, 1889): Is there evidence that $\Pr(\text{male}) = 0.5$?

Saxony families

Saxony families with 12 children having $k = 0, 1, \dots, 12$ sons.

k	0	1	2	3	4	5	6	7	8	9	10	11	12
n_k	3	24	104	286	670	1033	1343	1112	829	478	181	45	7



3

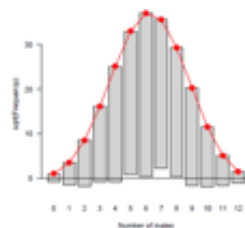
Common discrete distributions

Distribution	Counts, k	Values of X	$\Pr(X=k)$	Mean, $E(X)$	Var, $V(X)$
Bernoulli(p)	Success in 1 trial	$k=\{0, 1\}$	$p^k(1-p)^{1-k}$	p	$p(1-p)$
Binomial(n, p)	# successes in n trials	$0, 1, \dots, n$	$\binom{n}{k} p^k(1-p)^{n-k}$	np	$np(1-p)$
Geometric(p)	# of trials to 1 st success	$0, 1, 2, \dots$	$p(1-p)^k$	$\frac{1-p}{p}$	$\frac{1-p}{p^2}$
Neg. binomial(k, p)	# of trials to k^{th} success	$0, 1, 2, \dots$	$\binom{n+k-1}{k} p^k(1-p)^n$	$\frac{k(1-p)}{p}$	$\frac{k(1-p)}{p^2}$
Poisson(λ)	# of events in interval	$0, 1, 2, \dots$	$\frac{\lambda^k e^{-\lambda}}{k!}$	λ	λ
Log series(p)	# of types observed	$0, 1, 2, \dots$	$\frac{p^k}{n \log(1-p)}$		

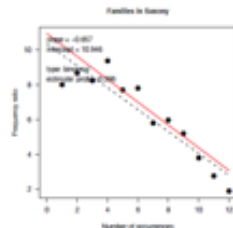
11

Graphing discrete distributions

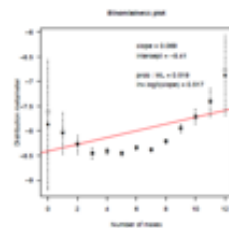
Rootgrams



Ord plots



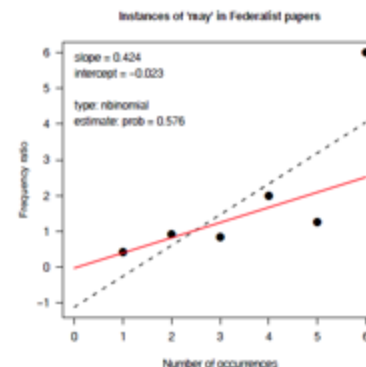
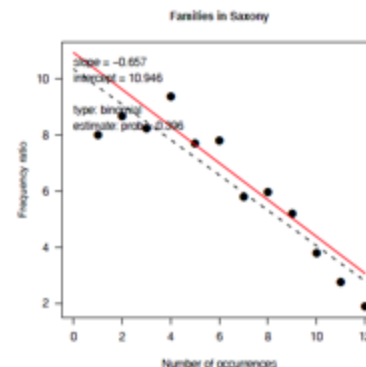
Robust distribution plots



Ord plots: Examples

Ord plots for the Saxony and Federalist data

```
> Ord_plot(Saxony, main = "Families in Saxony", gp=gpar(cex=1), pch=16)
> Ord_plot(Federalist, main = "Instances of 'may' in Federalist papers", gp=gpar(cex=1), pch=16)
```



03: Two-way tables

- Two-way tables summarize frequencies of two categorical factors
 - 2×2 : a special case, with **odds ratio** as a measure
 - $r \times c$: factors can be **unordered** or **ordered**
 - $r \times c \times k$: stratified tables, $r \times c$ with groups or circumstances
- Tests & measures of association
 - Pearson χ^2 , LR G^2 : **general association**
 - More powerful **CMH tests** for ordered factors
- Visualization
 - 2×2 : fourfold plots
 - $r \times c$: sieve diagrams, tile plots, ...
 - More graphical methods to come ...

Measures of association

• 2 × 2 tables

- Odds ratio

$$\theta = \frac{\text{odds}(B_1|A_1)}{\text{odds}(B_1|A_2)} = \frac{n_{11}/n_{12}}{n_{21}/n_{22}}$$

- Phi coefficient

- Analog of correlation
- $\phi^2 = \% \text{ of variance}$

$$\phi = \frac{n_{11}n_{22} - n_{12}n_{21}}{n_{1+}n_{2+}n_{+1}n_{+2}} = \pm \sqrt{\chi^2 / n}$$

• r × c tables

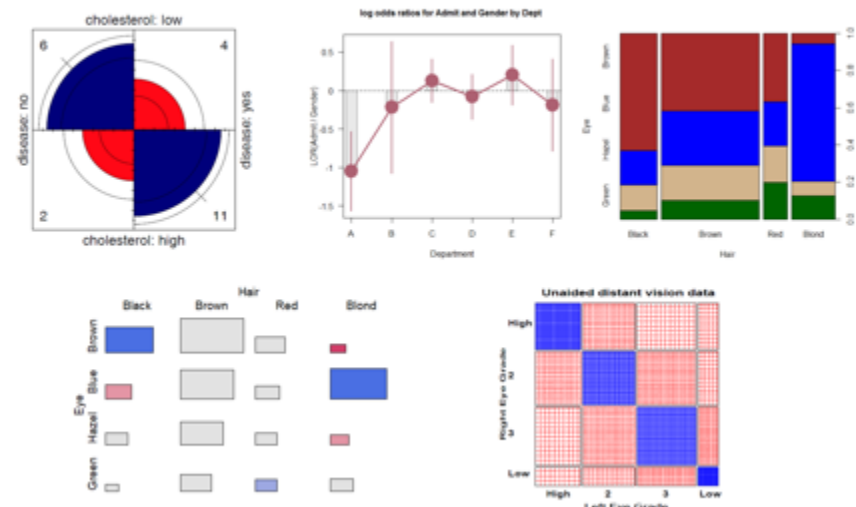
- Cramer's V – generalization of phi

$$\text{Cramer V} = \sqrt{\frac{\chi^2}{n \min(r-1, c-1)}}$$

- Pearson contingency coef

$$\text{Pearson C} = \sqrt{\frac{\chi^2}{\chi^2 + n}}$$

Visualizing association



CMH tests for ordinal factors

Three types of CMH tests:

Non-zero correlation

- Use when **both** row and column variables are ordinal.
- CMH $\chi^2 = (N-1)r^2$, assigning scores (1, 2, 3, ...)
- most powerful for **linear** association

Row/Col Mean Scores Differ

- Use when only **one** variable is ordinal
- Analogous to the Kruskal-Wallis non-parametric test (ANOVA on rank scores)

General Association

- Use when **both** row and column variables are nominal.
- Similar to overall Pearson χ^2 and Likelihood Ratio G^2 .

Observer agreement

- **Inter-observer agreement** often used as to assess reliability of a subjective classification or assessment procedure
 - → square table, Rater 1 x Rater 2
 - Levels: diagnostic categories (normal, mildly impaired, severely impaired)
- **Agreement vs. Association:** Ratings can be strongly associated without strong agreement
- **Marginal homogeneity:** Different frequencies of category use by raters affects measures of agreement
- **Measures of Agreement:**
 - Intraclass correlation: ANOVA framework— multiple raters!
 - Cohen's κ : compares the observed agreement, $P_o = \sum p_{ii}$, to agreement expected by chance if the two observer's ratings were independent, $P_c = \sum p_{i+} p_{+i}$.

$$\kappa = \frac{P_o - P_c}{1 - P_c}$$

04: Loglinear models, mosaic displays

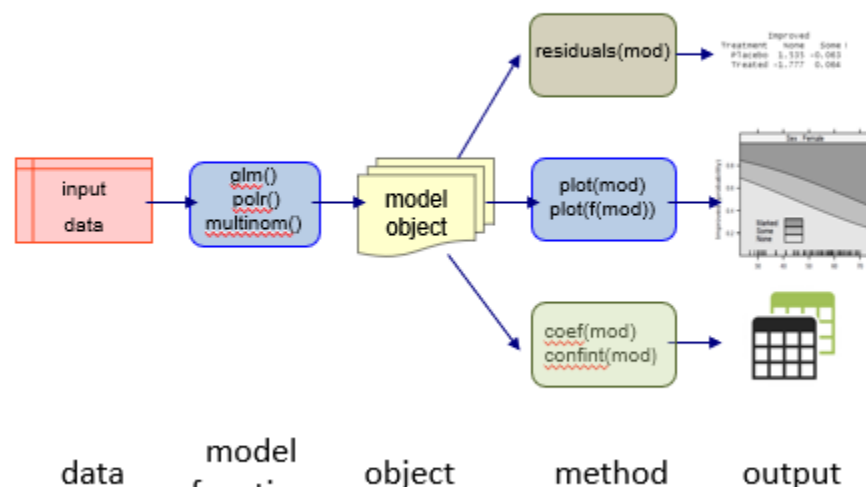
- Mosaic plots use sequential splits to show marginal and conditional frequencies in an n -way table
 - Shading: **sign** and **magnitude** of residuals $\rightarrow$ contributions to χ^2
 - Shows the pattern of association not accounted for
 - Permuting rows/cols often helps
- Loglinear models
 - Express associations with ANOVA-like interaction terms: $A*B, A*C$
 - Joint independence: $[AB][C] \equiv A * B + C$
 - Conditional independence: $[AC][BC] \equiv A \perp B \mid C$
 - Fitting models $\cong$ “cleaning the mosaic”
 - Response models: include all associations among predictors
- Sequential / partial plots & models
 - Sequential: Decompose all associations: $V_1; V_2|V_1; V_3|\{V_1, V_2\}, \dots$
 - Partial: Decompose conditional associations: $[V_1, V_2] \mid V_3 = \{a, b, \dots\}$

Loglinear models: Perspectives

Loglinear models grew up and developed from three different ideas and ways of thinking about notions of independence in frequency data

- **Loglinear approach:** analog of ANOVA; associations are ~ interactions
- **glm() approach:** analog of general regression model, for log(Freq), with Poisson distⁿ of errors
- **Logit models:** Loglinear, simplified for a **binary** response

Model-based methods: Fitting & graphing



Reduced models

- For a three-way table there is a range of models between mutual independence, $[A][B][C]$, and the saturated model, $[ABC]$
- Each model has an independence interpretation:
 $[A][B] \equiv A \perp B \equiv A \text{ independent of } B$
- Special names for various submodels

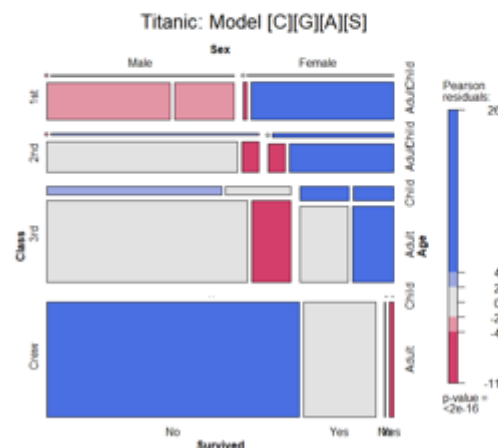
Table: Log-linear Models for Three-Way Tables

Model	Model symbol	Interpretation
Mutual independence	$[A][B][C]$	$A \perp B \perp C$
Joint independence	$[AB][C]$	$(A \ B) \perp C$
Conditional independence	$[AC][BC]$	$(A \ B) C$
All two-way associations	$[AB][AC][BC]$	homogeneous assoc.
Saturated model	$[ABC]$	ABC interaction

Fitting & visualizing models

```
mod0 <- loglm(~ 1 + 2 + 3 + 4, data=Titanic)
mosaic(mod0, main="Titanic: Model [C][G][A][S]")
```

In the model formulas, I'm using variable numbers 1-4 for **C**lass, **G**ender, **A**ge and **S**urvived



The **independence** model serves only as a background for the total associations in the table

Let's clean this mosaic!!

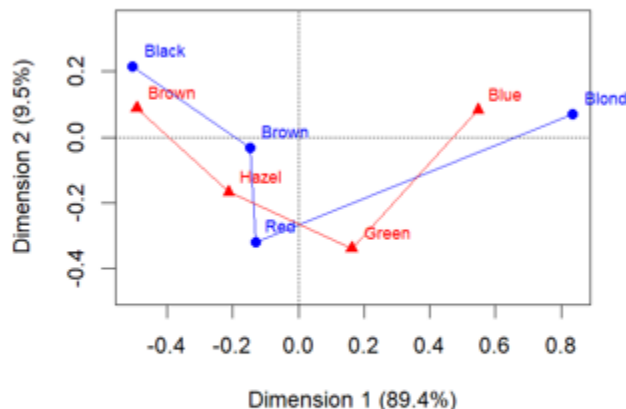
Note the scale of residuals:
+26 --- -11

05: Correspondence analysis

- CA is an exploratory method designed to account for association (Pearson χ^2) in a small number of dimensions
 - Row and column scores provide an **optimal scaling** of the category levels
 - Plots of these can suggest an explanation for association
- CA uses the **singular value decomposition** to approximate the matrix of residuals from independence
- Standard and principal coordinates have different geometric properties, but are essentially re-scalings of each other
- Multi-way tables can be handled by:
 - Stacking approach— collapse some dimensions interactively to a 2-way table
 - Each way of stacking $\rightarrow$ a loglinear model
 - MCA analyzes the full n – way table using an indicator matrix or the **Burt** matrix

Given a new 2-way table, my first thought is nearly always: `plot(ca(mytable))`

```
plot(haireye.ca, lines=TRUE)
```



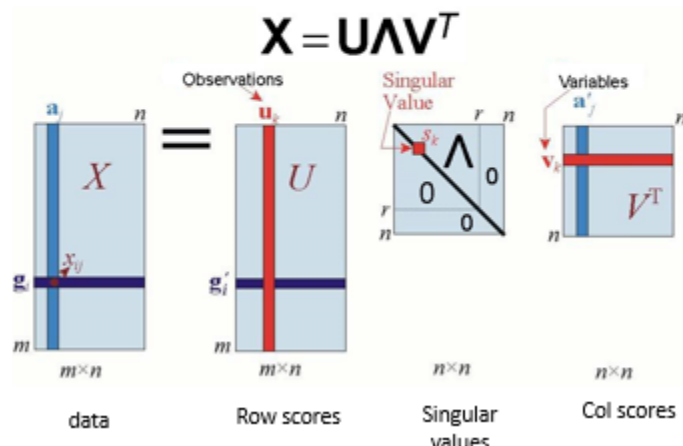
- Rough interpretation: row/col points “near” each other are positively associated (independence residuals $d_{ij} \gg 0$)
- Dim 1: 89.4% of χ^2 (dark $\rightarrow$ light)
- Dim 2: 9.5% of χ^2 (Red/Green vs. others)

6

Singular value decomposition

The singular value decomposition (SVD) is a basic technique for factoring a matrix and for matrix approximation

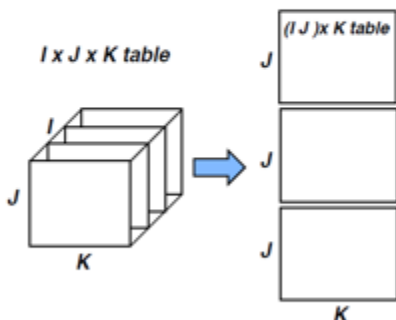
For an $m \times n$ matrix X of rank $r \leq \min(m, n)$ the SVD of X is:



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Multi-way tables: Stacking

A 3-way table of size $I \times J \times K$ can be sliced and stacked as a two-way table in several ways



- The variables combined are treated “interactively”
- Each way of stacking corresponds to a loglinear model
 - $(I \times J) \times K \rightarrow [AB][C]$
 - $I \times (J \times K) \rightarrow [A][BC]$
 - $J \times (I \times K) \rightarrow [B][AC]$
- Only the associations in separate $[]$ terms are analyzed and displayed
- The stacked table is analyzed with ordinary CA of the two-way stacked table

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Multiple correspondence analysis

- Extends CA to n -way tables
- Useful when simpler stacking approach doesn't work well, e.g., 10 categorical attitude items
- Analyzes all **pairwise bivariate** associations. Analogous to:
 - Correlation matrix (numbers)
 - Scatterplot matrix (graphs)
 - All pairwise χ^2 tests (numbers)
 - Mosaic matrix (graphs)
- Provides an **optimal scaling** of the category scores for each variable
- Can plot all factors in a single plot
- An extension, **joint correspondence analysis**, gives a better account of inertia for each dimension

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06: Logistic regression

- `loglm()` provides only overall tests of model fit
- Model-based methods, `glm()`, provide hypothesis tests, CIs & tests for individual terms
- Logistic regression: A `glm()` for a binary response
 - linear model for the log odds $\Pr(Y=1)$
 - All similar to classical ANOVA, regression models
- Plotting
 - Conditional, full-model plots show data and fits
 - Effect plots show predicted effects averaged over others
- Model diagnostics
 - Influence plots are often informative

Modeling approaches: Overview

Association models

- Loglinear models
(contingency table form)
[Admit][Gender Dept]
[Admit Dept][Gender Dept]
[AdmitDept][AdmitGender][GenderDept]
- Poisson GLMs
(Frequency data frame)
Freq ~ Admit + Gender * Dept
Freq ~ Admit*Dept + Gender*Dept
Freq ~ Admit*(Dept + Gender) + Gender*Dept
- Ordinal variables
Freq ~ right + left + Diag(right, left)
Freq ~ right + left + Symm(right, left)

Response models

- Binary response
 - Categorical predictors: logit models
logit(Admit) ~ 1
logit(Admit) ~ Dept
logit(Admit) ~ Dept + Gender
- Continuous/mixed predictors
 - Logistic regression models
Pr(Admit) ~ Dept + Gender + Age + GRE
- Polytomous response
 - Ordinal: proportional odds model
Improve ~ Age + Sex + Treatment
 - General multinomial model
WomenWork ~ Kids + HusbandIncome

4

Linear regression vs Logistic regression

OLS regression:

- Assume $y|x \sim N(0, \sigma^2)$

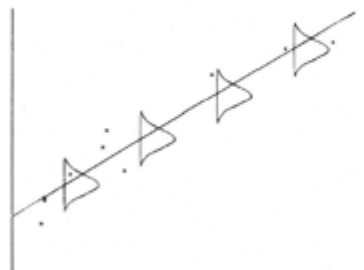


Fig. 2.1. Graphical representation of a simple linear normal regression.

y linear with x
constant residual variance

Logistic regression:

- Assume $\Pr(y=1|x) \sim \text{binomial}(p)$

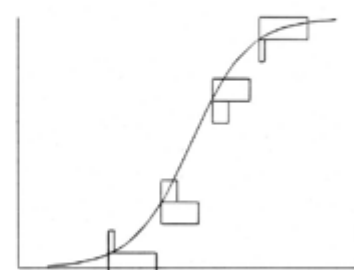


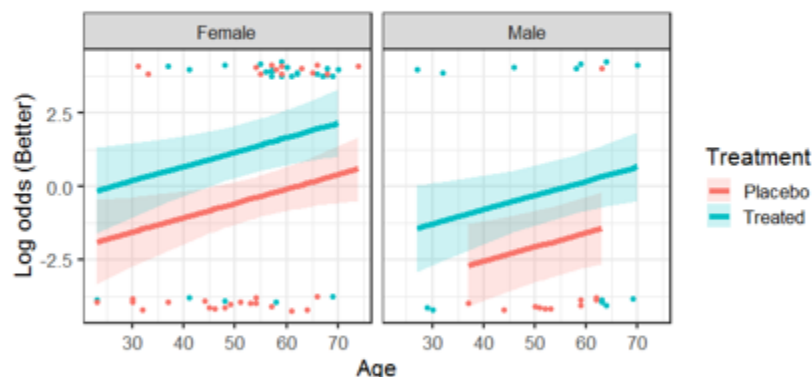
Fig. 2.2. Graphical representation of a simple linear logistic regression.

y ~ logit (x)
non-constant residual variance ~ p (1-p)

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Full-model plot

Plotting on the logit scale shows the additive effects of age, treatment and sex
NB: easier to compare the treatment groups within the same panel



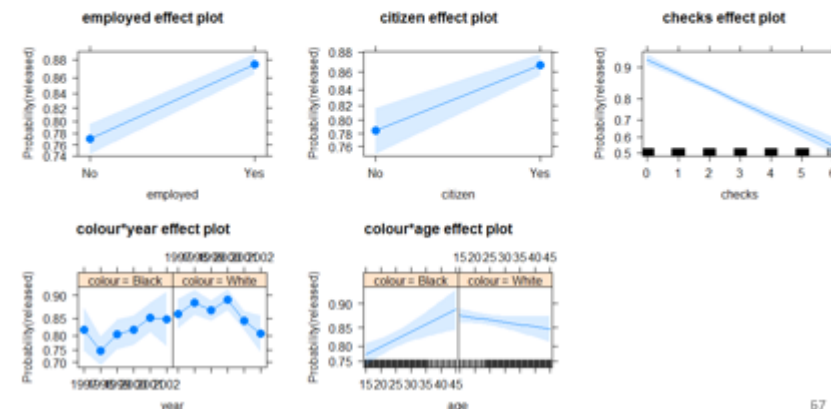
These plots show model uncertainty (confidence bands)
Jittered points show the data

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Effect plots: allEffects

All high-order terms can be viewed together using `plot(allEffects(mod))`

```
arrests.effects <- allEffects(arrests.mod,
xlevels=list(age=seq(15, 45, 5)))
plot(arrests.effects, ylab="Probability(released)", ...)
```



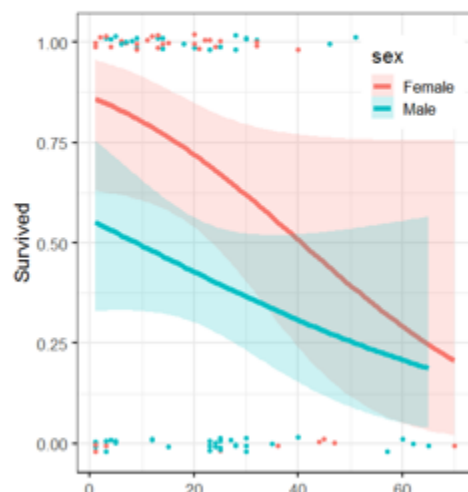
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07: Logistic regression: Extensions

- Polytomous responses
 - m response categories $\rightarrow (m-1)$ comparisons (logits)
 - Different models for **ordered** vs. **unordered** categories
- Proportional odds model
 - Simplest approach for ordered categories
 - Assumes same slopes for all logits
 - Fit with `MASS::polr()`
 - Test PO assumption with `VGAM::vglm()`
- Nested dichotomies
 - Applies to ordered or unordered categories
 - Fit $m - 1$ separate independent models $\rightarrow$ Additive G^2 values
- Multinomial logistic regression
 - Fit $m - 1$ logits as a single model
 - Results usually comparable to nested dichotomies, but diff interpretation
 - R: `nnet::multinom()`

Exploratory plots

Before fitting models, it is useful to explore the data with conditional ggplots



Survival decreases with age for both men and women

Women more likely to survive, particularly the young

Conf. bands show the data is thin at older ages

When response categories are

Ordered

Unordered

For example

No improvement
Some
Marked

NDP
Liberal
Conservative
Green

the analysis can use

Proportional odds model

Nested dichotomies

Multinomial logistic regression

we model these logits

None vs. Some or marked
None or Some vs. Marked

None vs. Some or marked
Some vs. Marked

NDP
Liberal
Green
Tory
Tory

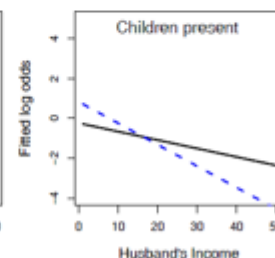
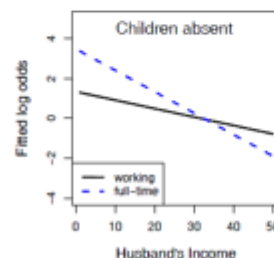
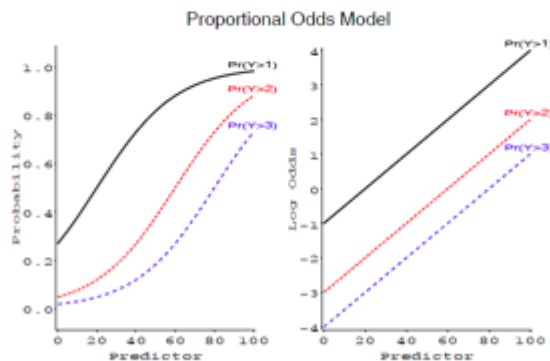
Nested dichotomies: Interpretation

Write out the predictions for the two logits, and compare coefficients:

$$\log \left(\frac{\Pr(\text{working})}{\Pr(\text{not working})} \right) = 1.336 - 0.042 \text{ H\$} - 1.576 \text{ kids}$$

$$\log \left(\frac{\Pr(\text{fulltime})}{\Pr(\text{parttime})} \right) = 3.478 - 0.107 \text{ H\$} - 2.652 \text{ kids}$$

Better yet, plot the predicted log odds for these equations:



08: Extending loglinear models

- Loglinear models, as originally formulated, were quite general, but treated all table variables as **unordered** factors
 - The GLM perspective is more general, allowing quantitative predictors and handling **ordinal factors**
 - The logit model give a simplified approach when one variable is a **response**
- Models for **ordered factors** give more powerful & focused tests
 - $L \times L$, R, C and R+C models **assign scores** to the factors
 - RC(1) and RC(2) models **estimate** the scores from the data
- Models for **square tables** allow testing structured questions
 - Quasi-independence: ignoring diagonals
 - symmetry & quasi-symmetry
 - theory-specific “topological” models
- These methods can be readily combined to analyze complex tables

Logit models

For a *binary* response, each loglinear model is equivalent to a logit model (logistic regression, with categorical predictors)

- e.g., Admit $\perp$ Gender | Dept (conditional independence $\equiv$ [AD][DG])

$$\log m_{ijk} = \mu + \lambda_i^A + \lambda_j^D + \lambda_k^G + \lambda_{ij}^{AD} + \lambda_{jk}^{DG}$$

So, for admitted ($i = 1$) and rejected ($i = 2$), we have:

$$\log m_{1jk} = \mu + \lambda_1^A + \lambda_j^D + \lambda_k^G + \lambda_{1j}^{AD} + \lambda_{jk}^{DG} \quad (1)$$

$$\log m_{2jk} = \mu + \lambda_2^A + \lambda_j^D + \lambda_k^G + \lambda_{2j}^{AD} + \lambda_{jk}^{DG} \quad (2)$$

Thus, subtracting (1)-(2), terms not involving Admit will cancel:

$$\begin{aligned} L_{jk} &= \log m_{1jk} - \log m_{2jk} = \log(m_{1jk}/m_{2jk}) = \text{log odds of admission} \\ &= (\lambda_1^A - \lambda_2^A) + (\lambda_{1j}^{AD} - \lambda_{2j}^{AD}) \\ &= \alpha + \beta_j^{\text{Dept}} \quad (\text{renaming terms}) \end{aligned}$$

where, α : overall log odds of admission; β_j^{Dept} : effect on admissions of department

Models for ordered categories

Consider an $R \times C$ table having *ordered* categories

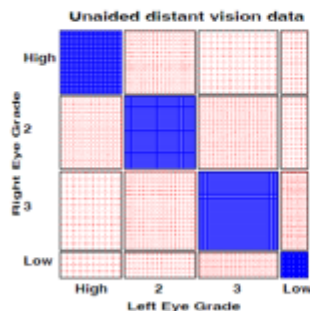
- In many cases, the *RC* association may be described more simply by assigning numeric scores to the row & column categories.
- For simplicity, we consider only integer scores, 1, 2, ... here
- These models are easily extended to stratified tables

R:C model	μ_{ij}^{RC}	df	Formula
Uniform association	$i \times j \times \gamma$	1	$i: j$
Row effects	$a_i \times j$	$(I - 1)$	$R: j$
Col effects	$i \times b_j$	$(J - 1)$	$i: C$
Row+Col eff	$ja_i + ib_j$	$I + J - 3$	$R: j + i: C$
RC(1)	$\phi_i \psi_j \times \gamma$	$I + J - 3$	Mult (R, C)
Unstructured (R:C)	μ_{ij}^{RC}	$(I - 1)(J - 1)$	$R: C$

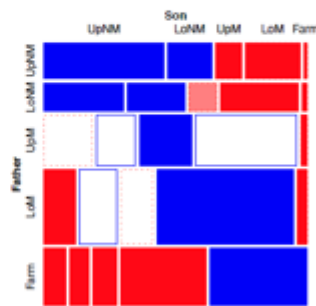
Square tables

Square tables arise when the row and column variables have the *same* categories, often *ordered*

Special loglinear models allow us to tease apart different *reasons* for association



Visual acuity data

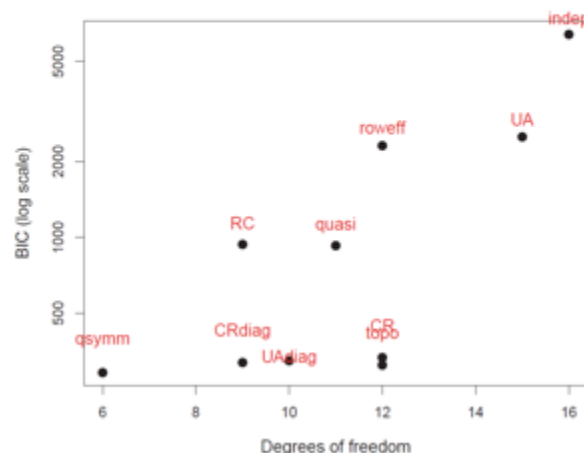


Hauser social mobility data

Model comparison plots

When there are more than a few models, a *model comparison plot* can show the trade-off between goodness-of-fit and parsimony

- This sorts the models by both *fit* & *complexity*



Plot BIC vs. df

Can also use AIC, or G^2 / df in this plot

Plot on log scale to emphasize *diffs* among better models

And, the winner is: Quasi-symmetry!

09: GLMs for Count Data

- GLMs provide a unified framework for linear models
 - Different families, all estimated in the same way
 - →link function and associated variance function
- For count data, starting from $\log(\mu) = \mathbf{X} \beta$, $\mu | \mathbf{X} \sim$ Poisson:
 - Overdispersion → quasi-poisson, negative binomial
 - Standard tools for assessing model fit
- Excess zero counts introduce new ideas & methods
 - ZIP model: structural model for the 0s
 - Hurdle model: random model for 0s, 2nd model for $Y > 0$
- In all this, we rely on data & model **plots** for understanding

Canonical links and variance functions

- For every distribution family, there is a default, **canonical link** function
- Each one also specifies the expected relation between the mean and **variance**

Table 11.2: Common distributions in the exponential family used with generalized linear models and their canonical link and variance functions

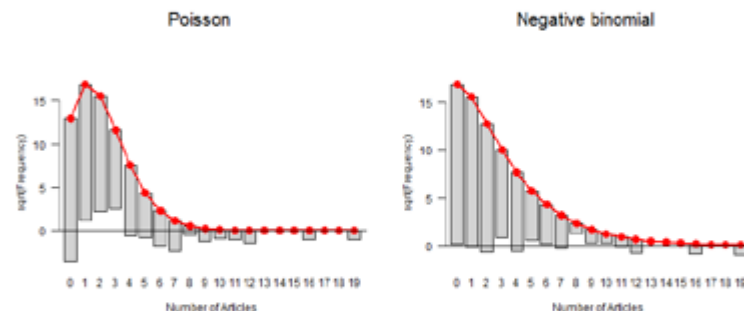
Family	Notation	Canonical link	Range of y	Variance function, $V(\mu \eta)$
Gaussian	$N(\mu, \sigma^2)$	identity: μ	$(-\infty, +\infty)$	ϕ
Poisson	$\text{Pois}(\mu)$	$\log_e(\mu)$	$0, 1, \dots, \infty$	μ
Negative-Binomial	$\text{NBin}(\mu, \theta)$	$\log_e(\mu)$	$0, 1, \dots, \infty$	$\mu + \mu^2/\theta$
Binomial	$\text{Bin}(n, \mu)/n$	$\text{logit}(\mu)$	$\{0, 1, \dots, n\}/n$	$\mu(1 - \mu)/n$
Gamma	$G(\mu, \nu)$	μ^{-1}	$(0, +\infty)$	$\phi\mu^2$
Inverse-Gaussian	$IG(\mu, \nu)$	μ^2	$(0, +\infty)$	$\phi\mu^3$

Choose a basic family:

- Get a default, canonical link, $g(\mu)$
- Also get a variance function for free!

First, look at rootograms:

```
plot(goodfit(PhdPubs$articles), xlab = "Number of Articles",
     main = "Poisson")
plot(goodfit(PhdPubs$articles, type = "nbinomial"),
     xlab = "Number of Articles", main = "Negative binomial")
```



One reason the Poisson doesn't fit: excess 0s (some never published?)

Q: What might some other reasons be?

Think back to assumptions: independent obs; constant probs; unmodelled vars

Quasi-poisson models

- The **quasi-poisson** model allows the dispersion, ϕ , to be a free parameter, estimates with other coefficients
- The conditional variance is allowed to be a multiple of the mean

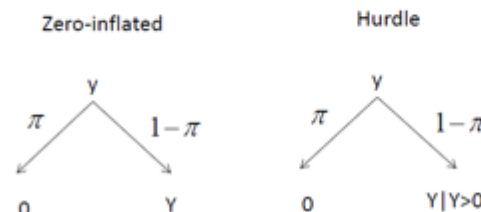
$$\text{Var}(y_i | \eta_i) = \phi \mu_i$$

- This model is fit with `glm()` using **family=quasipoisson**
 - The estimated coefficients $\hat{\beta}$ are **unchanged**
 - The standard errors are multiplied by $\phi^{1/2}$
 - Peace, order & good government is restored!

Models for excess zeros

Two types of models, with different mechanisms for zero counts

- zero-inflated models:** The responses with $y_i = 0$ arise from a mixture of **structural**, always 0 values, with $\text{Pr}(y_i = 0) = \pi$ and the rest, which are **random** 0s, with $\text{Pr}(y_i = 0) = 1 - \pi$
- hurdle models:** One process determines whether $y_i = 0$ with $\text{Pr}(y_i = 0) = \pi$. A second process determines the distribution of values of positive counts, $\text{Pr}(y_i | y_i > 0)$



10: Models for log odds & LORs

- Logit models for a binary response generalize readily to a polytomous response
 - → Models for log odds, familiar interpretation
 - Handles 3+ way table, ordinal variables
 - Simple plots for interpretation
- Generalized odds ratios handle bivariate responses
 - Simple linear models for LOR
 - Easy to model log odds for each response and the LOR simultaneously
 - Easy to visualize results

Main ideas

- Familiar case— Binary responses:
 - Every loglinear model for a binary response has an equivalent form in terms of **log odds** ["logit" models]
 - Log odds models have simple interpretations
 - Data + model plots give simple descriptions of data and models
- Extend to two-way ($I \times J$) and three-way + ($I \times J \times K_1 \dots$) tables:
 - Log odds as **contrasts** in $\log(n)$
 - Variety of simple models for log odds (ANOVA-like)
 - Easily incorporate **ordinal** variables
 - Data + model plots give simple descriptions of data and models
- Generalized log odds ratios capture associations between two **focal variables**
 - Simple linear models for LOR
 - Direct visualization (Data + model plots) $\Rightarrow$ more sensitive comparisons

Based on my CARME (2015) presentation, <https://www.datavis.ca/papers/CARME2015-2x2.pdf>

3

Models for log odds

A variety of simple models can be specified in terms of log odds:

Table: Models for adjacent log odds in an $I \times J$ table with B as the response

Model	log odds parameters	degrees of freedom
null log odds	$\psi_{ij}^{AB} = 0$	$I(J - 1)$
constant log odds	$\psi_{ij}^{AB} = \psi$	$I(J - 1) - 1$
uniform B log odds	$\psi_{ij}^{AB} = \psi_i^A$	$I(J - 2)$
parallel log odds	$\psi_{ij}^{AB} = \psi_i^A + \psi_j^B$	$(I - 1)(J - 2)$
saturated	ψ_{ij}^{AB} unspecified	

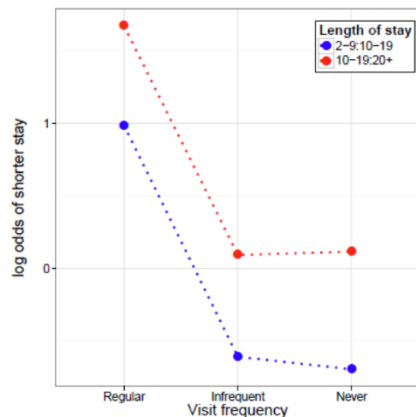
- The log odds, ψ_{ij}^{AB} can be viewed as entries in an $I \times (J - 1)$ table
- These models are analogous to ANOVA tests of the A, B and $A * B$ effects in this table.

These are similar to some `glm()` models we looked at before, but here in a simpler way by thinking in terms of log odds

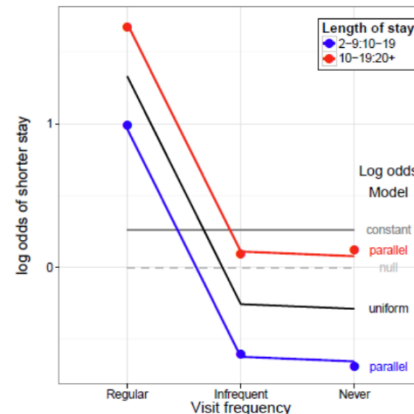
13

Visualizing log odds and models

Plots of observed and fitted log odds: easy interpretation of data and models



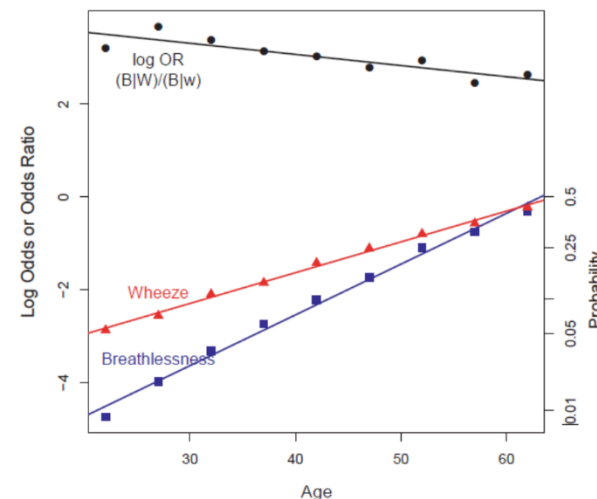
Data plot: Observed log odds



Data + Model plot (fitted log odds)

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Linear model for log odds and log odds ratios

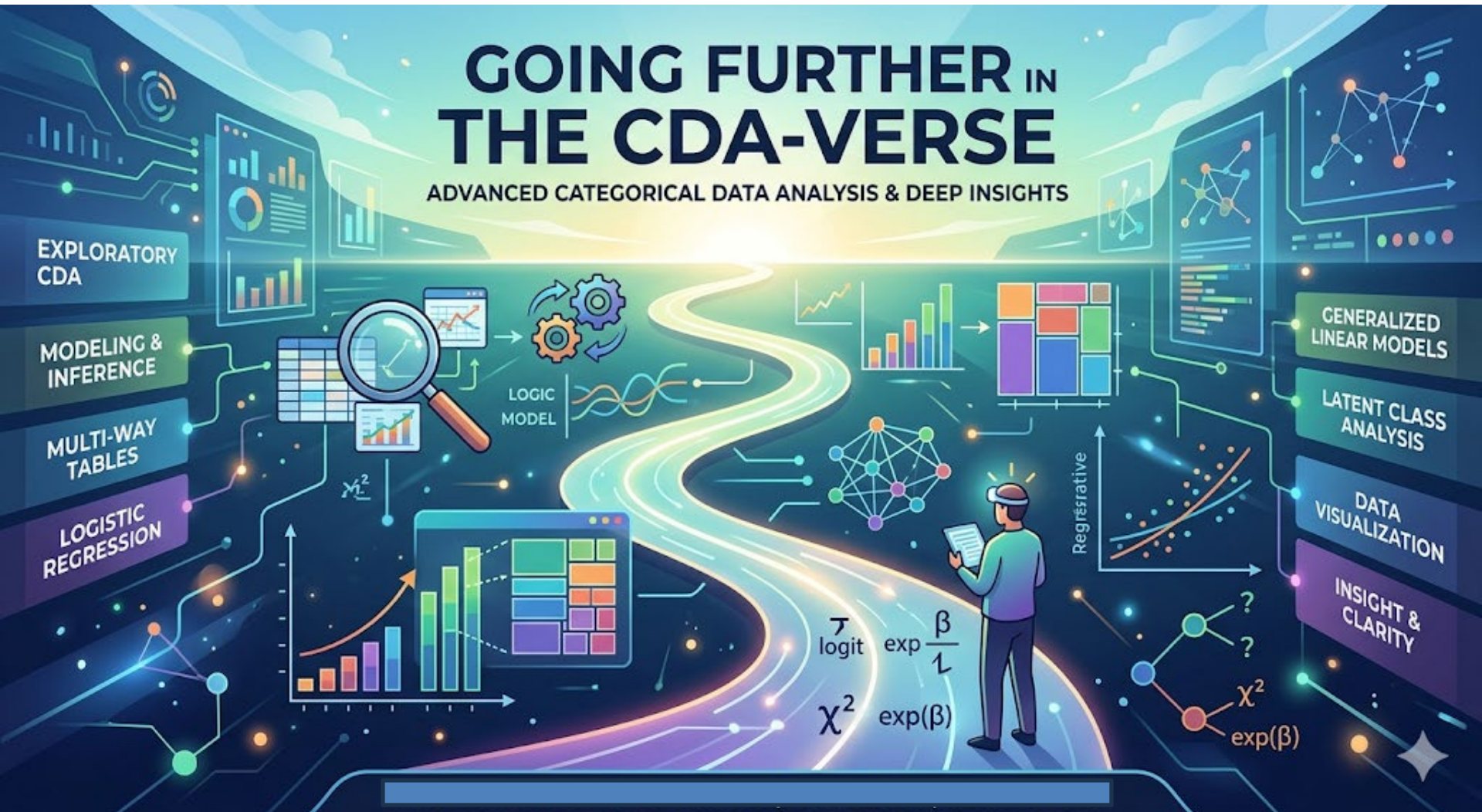


Log odds & LORs have similar scales, so it is not terrible to plot them together

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GOING FURTHER IN THE CDA-VERSE

ADVANCED CATEGORICAL DATA ANALYSIS & DEEP INSIGHTS



Going further: What did I leave out?

- This course, like DDAR, takes an entirely frequentist approach
 - Treats model parameters as **fixed**, constants
 - p -values, CIs interpreted as “over many repeated samples ...” --- 95% would contain the true value
 - Never really know if your sample gives correct inferences
- Bayesian approach
 - Treat parameters as random variables, incorporating **prior beliefs** and updating them with observed data to produce a **posterior distribution**. Much more intuitive.
 - A Bayesian 95% **credible interval** **directly** states there is a 95% probability the parameter lies within that range



Fisher vs. Bayes



- Small samples, sparse data
 - Frequentist methods may fail to converge, or require fixups (add 0.5 to avoid 0 cells) for χ^2 tests to be valid
 - Bayesian approaches use priors to "smooth" empty cells, $\Rightarrow$ more robust estimates in high-dimensional tables
- Logistic regression
 - Frequentist: relies on large-sample asymptotics for inference
 - Bayesian: provides full posterior distⁿ for odds ratios, β s
- Computation
 - Frequentist: very fast, particularly for large datasets
 - Bayesian:
 - Requires Bayesian sampling, e.g., MCMC; did my chains converge??
 - Originally (in DDAR): software quite limited, awkward to use
 - Now, much easier to use.

Bayesian advances

- Regularization methods: stabilize inference with small or zero counts
- New variable selection methods
 - Effect clustering: merge levels of categorical predictor that have the same effect $\Rightarrow$ more stable in high-D problems
- Applications:
 - Clinical trials: Bayesian adaptive designs can use historical data to speed up decision on efficacy of drug or treatment
 - Epidemiology: Bayesian hierarchical models used in disease mapping (estimating risk across areas), pooling strength

PERSPECTIVE

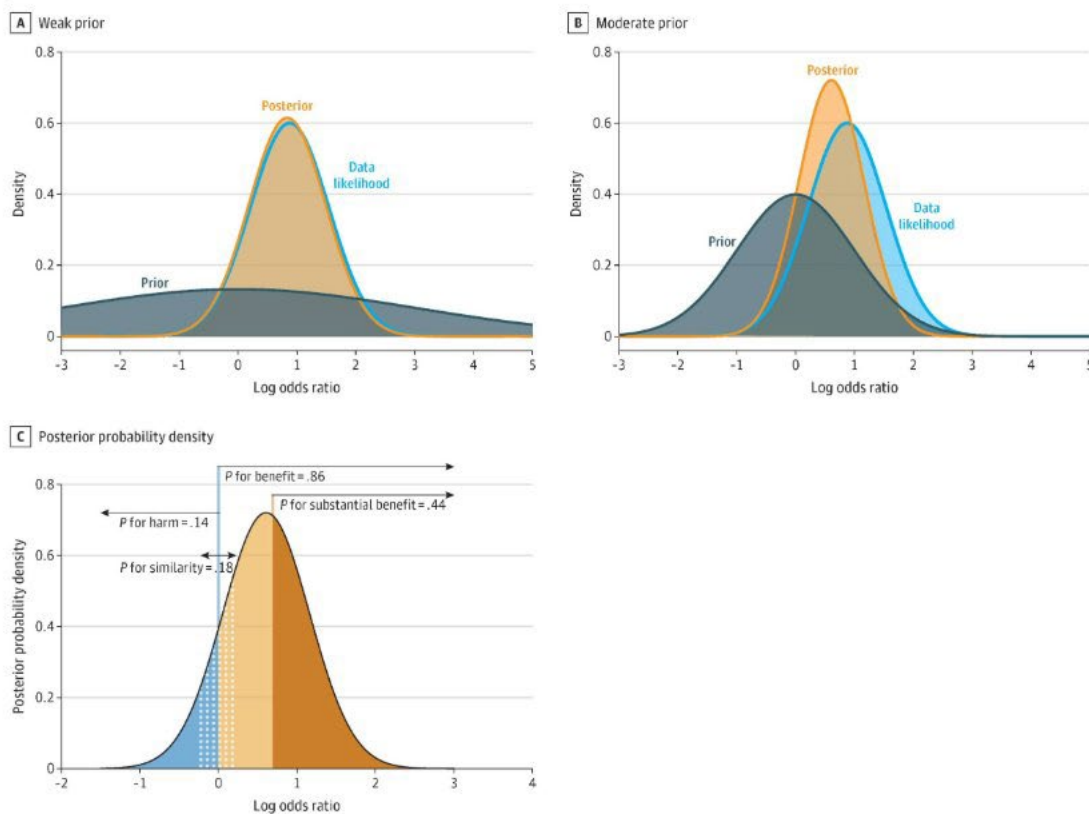
Embracing Bayesian Methods in Clinical Trials FDA's Long-Awaited Draft Guidance

J. Jack Lee, PhD; Frank E. Harrell Jr, PhD; Lisa M. LaVange, PhD; David J. Spiegelhalter, PhD

BREAKING NEWS: Bayesian methods adopted in JAMA recommendations to FDA

<https://t.co/7mMecV2E11>

Figure. Distribution Plots for Log Odds Ratios (ORs)



Hierarchical, longitudinal data

- All methods in this course — χ^2 tests, loglinear models, logistic regression, ...— assume observations are **independent**
- Fails when:
 - **same person** is measured at multiple time points (longitudinal / repeated measures), or
 - people are **nested within groups** such as schools, clinics, or geographic regions (hierarchical).
- Ignoring dependence:
 - Underestimated std. errors: CIs too narrow, p -values too small ( phantom precision)
 - Biased parameter estimates

Longitudinal CDA designs

Data structure: n subjects, each measured at T occasions.

- Categorical response: — e.g., disease present/absent, symptom severity (ordered), count of events in a period.
- Examples:
 - Panel surveys: political preference over election cycles
 - Clinical: patients worse, stable, improved at 4, 8, 12 weeks
 - Epidemiology: hospital visits per patient over 3 years

Hierarchical (clustered / multilevel) designs

Data structure: Individuals at level 1 are nested within groups at level 2 (maybe: level 3, ...).

- The grouping creates dependence: two students in same school, or two patients at same clinic, share unmeasured **contextual influences**.
- Examples:
 - Students (level 1) nested in classrooms (level 2) nested in schools (level 3); binary outcome = passed a standardized test.
 - Patients nested in hospitals; ordinal outcome = satisfaction rating.
- Why standard methods fail:
 - Observations within the same cluster are positively correlated (intraclass correlation, ICC)
 - Std. logistic / Poisson regression treats all as independent, inflating the effective sample size, giving over-confident inference
 - Group-level predictors tested against wrong error term.

Modeling approaches: GEE

- Marginal models via Generalized Estimating Equations (GEE)
 - Fit a population-average GLM (logistic, Poisson, ordinal, ...) as usual, but explicitly model the within-S correlation structure (exchangeable, AR(1), unstructured, ...).
 - GEE adjusts standard errors so inference is valid even if the correlation structure is mis-specified (robust/“sandwich” SEs).
 - Interpretation: coefficients describe the **average effect in the population**, not effects for a specific individual.
- R: `geepack::geeglm()`, `gee::gee()`

General Linear Mixed Models (GLMM)

The natural extension of mixed models to non-normal outcomes. For a binary response with two-level nesting:

$$\text{logit}[P(Y_{ij} = 1)] = \mathbf{x}_{ij}^T \boldsymbol{\beta} + u_j$$

where $u_j \sim N(0, \sigma_u^2)$ is a random intercept for group j .

- $\boldsymbol{\beta}$: fixed effects (individual- and group-level predictors).
- σ_u^2 : variance of group-level random effects
- Can add **random slopes** if effect of predictor varies across groups
- R: `lme4::glmer()`, `glmmTMB::glmmTMB()`, `brms::brm()` (Bayesian)

Transition (Markov) models

- Model response at time t as a function of covariates *and* the lagged response $Y_{i,t-1}$ (or several lags).
- Directly captures how **prior** state predicts current state
- Natural for panel /clinical data where change over time is the scientific focus.
- Equivalent to a conditional logistic regression on each transition.

Connections to course topics

Course topic	Longitudinal/hierarchical extension
Logistic regression (binary)	GLMM with logit link; GEE with binary family
Ordinal logistic (proportional odds)	Mixed proportional-odds model (<code>ordinal::clmm()</code>)
Poisson / negative binomial (counts)	Poisson GLMM; NB GLMM (<code>glmmTMB</code>)
Multinomial logistic	Mixed multinomial (<code>mclogit::mblogit()</code>)
Loglinear models	Mixed log-linear for repeated cross-classifications

R software summary

Package	Primary use
<code>lme4</code>	GLMMs: <code>glmer()</code> for binary/count; fast, widely used
<code>glmmTMB</code>	GLMMs with more families (NB, zero-inflated, ordinal), random slopes
<code>geepack</code>	GEE: <code>geeglm()</code> ; marginal models, robust SEs
<code>ordinal</code>	Mixed ordinal models: <code>clmm()</code>
<code>brms</code>	Bayesian GLMMs via Stan; full posterior for all variance components

WHAT! There's even more??

Five other areas in a totally comprehensive CDA course:

1. Latent Class Analysis (LCA)
2. Survival Analysis / Event History Models
3. Item Response Theory (IRT)
4. Missing Data & Multiple Imputation
5. Regularized Logistic Regression

Latent Class Analysis & Survival Analysis

Latent Class Analysis (LCA)

- Factor analysis for categorical indicators: finds latent subgroups explaining associations among observed variables
- Widely used for diagnostic categories, behavioral typologies, health risk profiles
- R: `poLCA`, `mclust`
- Agresti Ch. 13; Collins & Lanza (2010), *Latent Class and Latent Transition Analysis*

Survival Analysis / Event History Models

- Outcome is *time to event* (relapse, death, dropout, recovery) — possibly censored
- Workhorses: discrete-time logistic regression; Cox proportional hazards
- Very common in clinical and health psychology
- R: `survival`, `flexsurv`; Agresti Ch. 12; Hosmer, Lemeshow & May (2008)

Item Response Theory & Missing Data

Item Response Theory (IRT)

- Models P (correct or endorsed response) as a function of a latent trait (ability, attitude)
- Rasch, 2PL, 3PL models are logistic regression with a latent predictor
- Natural extension of GLMs — directly relevant to psychometrics
- R: `ltm`, `mirt`, `TAM`; Embretson & Reise (2000)

Missing Data / Multiple Imputation

- Cannot mean-impute a categorical variable; complete-case analysis distorts association estimates
- MICE (multiple imputation via chained equations) handles categorical predictors and outcomes directly
- R: `mice`; van Buuren (2018), *Flexible Imputation of Missing Data* (free online)

Regularized Logistic Regression

The problem: when ordinary logistic regression fails

- Many predictors $\rightarrow$ instability, perfect separation, near-collinearity
- LASSO (L_1) shrinks coefficients to zero $\rightarrow$ automatic variable selection
- Ridge (L_2) shrinks toward zero $\rightarrow$ stable estimation, all predictors retained
- Elastic net combines both penalties

Tools and further reading

- Frequentist complement to Bayesian regularization (shrinkage priors)
- R: `glmnet`
- Hastie, Tibshirani & Friedman (2009), *The Elements of Statistical Learning*, Ch. 4, 18

CDA Dad Jokes

- **Q:** What did the χ^2 statistic say to her p -value?
 - A: You're the only one who can tell me if our relationship is real
- **Q:** Why was the mosaic plot so calm?
 - A: He had already made peace with all his residuals.
- **Q:** Why did logistic regression break up with his linear regression?
 - A: It's not you honey, it's me! My love is binary — I can only give you 0 or 1.
- A loglinear model walks into a bar.
 - Bartender says: "We don't serve your type here."
 - Model says: "Fine — I'll just fit a main effects model and assume no one here is associated."



Your turn: Feedback?

What did you like/dislike about 6136?

- Topics: what were the:
 - most interesting?
 - most boring?
 - most challenging/difficult?
- How useful were the web materials for the course?
 - Quizzes? R scripts? Resources / extra readings?
- What topics/activities did you learn most from?
- What gave you the most difficulty?
- How does this relate to your own work?

Tips for next time ...

- What should I try to differently the next time?
 - More of X?
 - Less of Y?
 - Aspects of how the course is structured?
 - Evaluation?

